

Algorithmic Gatekeeping and Agenda-Setting in Online News Discussions

Anonymous submission

Abstract

Ranking policies in online news comment sections determine which contributions are presented to readers first. Prior research examines the attributes of curated comments and the effects of ranking social-media feeds, but the combined effects of algorithmic ordering, editorial pinning, and reply handling on the representation of a finite discussion remain less understood. Using a corpus of 2.39 million comments on 5,118 *Der Standard* articles published in 2025, we evaluate the effects of 90 ranking algorithms on the kinds of comments that are prioritised in discussions. We introduce the Feature-Oriented Ranking Utility Metric (FORUM) to measure how policies promote or suppress comment attributes relative to random, best-possible, and worst-possible feature orderings. We complement this with topic modelling to compare the presented news topic agenda with the article and an audience-vote-derived reference. Predictive comment ranking models tend to prioritise deliberative quality, article similarity, and lexical diversity, and suppress novelty, while reply handling can offset these effects. Early prominence also differs from full-discussion presentation. Predictive models produce larger article-directed than reader-directed topical agenda gains, and hiding replies increases article alignment. These results show that editorial influence extends beyond pinned selections to the algorithms and interfaces that organise discussion. Making these choices explicit can help newsrooms assess trade-offs between topical focus, comment attributes, and the representation of reader contributions.

Code —

<https://anonymous.4open.science/r/commentgap-15CB>

Introduction

An online news article may attract hundreds of comments, but their ordering gives some contributions greater prominence than others. Which contributions appear depends on more than what readers write. Ordering by votes, posting time, ‘recommended’, pinning Editors’ Picks, and displaying or hiding replies all determine which perspectives receive prominence. The same discussion organised in different ways can therefore present different accounts of readers’ responses to a story.

These choices extend editorial influence beyond the article itself. They involve algorithmic gatekeeping through the allocation of comment visibility and agenda-setting through

the organisation of issue and attribute salience. Editors can manually highlight individual contributions, while ranking algorithms are responsible for distributing visibility across the remaining discussion. Reader votes supply another selection signal, but their influence depends on how the interface uses them. Prior work examines the characteristics of selected comments and the consequences of comment curation (Diakopoulos 2015; Juarez Miro 2022; He et al. 2024; Wang and Diakopoulos 2022), but less is known about how ordering, pinning, and reply handling jointly shape the attributes and topics presented from the same comment pool. We examine these choices together, asking both what kinds of contributions they bring forward and whose topical priorities the resulting presentation resembles.

We focus our attention on two research questions:

- RQ1 How can algorithmic ordering, reply handling, and comment pinning change the attributes surfaced by a discussion, and how do these effects depend on the depth of the visible comments?
- RQ2 How can algorithmic ranking shift the visible topic distribution towards or away from journalist vs audience-derived agendas?

To address these, we gather 2.39 million comments across 5,118 articles from 2025 from Austrian newspaper *Der Standard*, one of the largest German-speaking news forums. We compare ranking algorithms across 90 combinations of ordering, reply presentation, and comment pinning. For RQ1, we develop the “Feature-Oriented Ranking Utility Metric” (FORUM) to measure which comment attributes are visible earlier under different ranking algorithms, relative to random, best-possible, and worst-possible feature orderings. With RQ2, we fit an article–comment topic model with BERTopic, and assess how different ranking algorithms represent the topics present in the discussion relative to the article and expressed reader preferences.

Related Work

Algorithmic gatekeeping and agenda-setting

Gatekeeping concerns the selection and access decisions through which some contributions reach public communication while others receive little or no visibility. Early work conceptualised this as journalists’ selection of news events (White 1950); later accounts describe the broader process

through which information passes, or is closed off from, media attention (Shoemaker and Vos 2009). In digital settings, journalists, users, and algorithms all participate in selection through platforms with different mechanisms (Wallace 2018). Audience engagement can itself act as a gatekeeping signal, while “gatewatching” describes the collaborative monitoring and highlighting of already available information (Shoemaker and Vos 2009; Bruns 2003). On the Internet, gatekeeping power now extends to popular search engines and social networks (Nielsen 2016). Audits of news curation likewise reveal systematic differences in the source diversity and topic composition produced by algorithmic and editorial selection (Bandy and Diakopoulos 2020).

Agenda-setting theory, first developed by McCombs and Shaw (1972), posits that mass media influence public opinion not by dictating what to think, but by making certain issues more salient than others in shaping what people think about. Building on this, second-level agenda-setting theory shifts the focus from issue salience to attribute salience—how media coverage influences the public’s understanding of how to think about an issue, including its tone, framing, and associations (McCombs et al. 1997). Beyond first- and second-level effects, extensions examine networks of associated issues, influence between media, and how audiences shape and combine agendas in their information environments (Guo and McCombs 2011; Atwater, Fico, and Pizante 1987; Kim and Lee 2006; Shaw et al. 1999).

In the digital age, algorithmic systems have transformed news production and consumption, prompting a reconceptualisation of classic media theories and empirical studies (Van Dijck and Poell 2013; Voinea 2025). Platforms now act as “algorithmic gatekeepers”—selecting, ranking, and personalising news and opinion pieces (Wallace 2018). Increased attention is now being paid to their agenda-setting consequences. For instance, Trielli and Diakopoulos (2022) show that Google search results during the 2020 U.S. election exhibited differences in topic salience compared to the news media agenda, with limited user power to reshape these. In a controlled experiment, Einarsson, Helles, and Lomborg (2025) find that personalised news recommendations subtly narrow the range of topics towards softer news in terms of both user exposure and consumption. Such studies demonstrate how ranking systems across platforms help shape what is emphasised in public discourse. We extend this perspective to comment sections: by analysing how ranking policies prioritise certain types of comments, we examine a form of algorithmic gatekeeping and agenda-setting in public discourse.

Ranking and the governance of discussion

Users often lack the time to scan hundreds of comments in online discussions, missing potentially interesting contributions and sometimes ending their participation altogether (Levy 2008; Lampe et al. 2014). On some platforms, users view only 3% of comments (Althuniya et al. 2020). Beyond news comments, Piccardi et al. (2025) show in a field experiment on X that re-ranking posts changes exposure and affective polarisation, while audits comparing algorithmic and chronological feeds find differences in account visibil-

ity (Efstratiou et al. 2026). Feed audits also associate rank with substantial differences in subsequent activity (Chan et al. 2026). Such findings motivate examining ranking interventions, while our analysis concerns presentation rather than downstream attitudes. Within comment sections, an exploratory audit has also found user-to-user variation in visible comment rankings, motivating more systematic study of how ranking shapes discourse (Nutakki, Kempermann, and Weber 2026). Automated assessment and ranking are increasingly important (Momeni, Cardie, and Diakopoulos 2015). For example, the German language focused AQuA (Behrendt et al. 2024) combines predictions for 20 dimensions into a weighted deliberative quality score informed by expert and non-expert judgements. Ideally, users see a summary of the discussion through selected comments, though this is challenging without biasing the results. Many news organisations rank comments in reverse-chronological order by default, or allow sorting by user up/downvotes. Some run an *Editors’ Picks* section, where selected comments are made prominently visible to readers.

Defaults matter because most users do not change comment sorting. Comment sorting can foster participation and constructive discussion (Diakopoulos and Naaman 2011), while highlighting high-quality contributions may improve commenters’ subsequent contributions (Wang and Diakopoulos 2022). Negative evaluations can increase posting while lowering later quality (Cheng, Danescu-Niculescu-Mizil, and Leskovec 2014), and community responses to incivility depend on conditions supporting collective corrective action (Ziegele, Naab, and Jost 2020). Concentrated participation may also produce a more hostile visible tone (Kim and Noh 2026).

Comment ranking and feedback influence which comments users read as well as write. Editor-picked comments provide a debate overview but may reflect undisclosed editorial policies. Community-based ranking can sanction incivility but may discourage minority opinions. This creates tension between audience participation and journalistic gatekeeping (Masullo Chen and Pain 2017; Diakopoulos 2015). Users’ and editors’ judgements often differ. Juarez Miro (2022) discusses the “comment gap”—only 17.2% of New York Times comments made both the readers’ and Editors’ Picks lists, as readers favour direct, confrontational, and aligned comments, while editors chose more articulate, conciliatory, and diverse ones. Using approximately 80,000 New York Times articles, He et al. (2024) find that Editors’ Picks tend to be longer, more article-relevant, more positive, and less toxic than other comments, with selection criteria varying across news sections. Whilst the majority of this closely related work is in English, efforts to expand analysis through data work have also emerged in German, such as Fraxanet et al. (2026)’s *Der Standard* 2013–2022 forum archive. Together, these studies motivate examining further how different algorithmic factors jointly shape the attributes and topics made prominent within a fixed discussion.

Data and Methods

Data collection

Der Standard is one of Austria’s two most-read “quality-newspapers” (“qualitätszeitung”, roughly equivalent to a “broadsheet”), hosting one of the largest German-speaking online forums globally with 73,000 active users annually (Burger 2022). Our analysis uses the publicly accessible articles and comments from stories in the year 2025. Reading is publicly available, while commenting and voting require registration (pseudonymously, if desired). Pinned posts (or “Editors’ Picks”) in discussions are comments deemed especially valuable by editors, community management, or community guides¹. They are shown at the top of the discussion and marked as *pinned* (“Angeheftet”).

Our data collection uses the same underlying corpus and collection pipeline as the companion study (Anonymous 2026). We programmatically collected accessible articles published in 2025 and their public comment records using monthly sitemaps and the site’s public GraphQL endpoint, retaining comment text, hashed author usernames, comment relationships, posting timestamps, vote totals, and pinned status, alongside article text and metadata. The initial collection contains 9,415,942 comment records, including 193,090 deleted-comment tombstones. We also collected December 2024 records for author-history features, excluding them from the 2025 outcome sample. We focus our analysis on discussions containing at least one comment selected as Editors’ Pick and at least one non-selected comment. From 44,596 retrieved articles, the final sample contains 5,118 discussions with 2,392,952 candidate comments, of which 20,930 are Editors’ Picks. The mean number of comments per discussion is 467.6 (median 325), and the mean number of Editors’ Picks is 4.09 (median four). Combining up- and downvotes, the mean total is 7.06 votes per candidate comment (median two). We use one common article split for all selector models: 2,559 development and 2,559 held-out articles, stratified by publication month and joint candidate-size tercile. The policy and topic analyses, as the focus of this paper, use the held-out partition and additionally require more than 10 comments in a discussion, leaving 2,556 discussions and 1,203,611 comments.

Constructing ranking algorithms

We define different possible comment ranking policies. Each comment ranking algorithm is made up of three policy elements: primary ordering, Editors’ Pick status, and reply structure. Primary ordering is the main way in which comments are ranked relative to each other (e.g. in order of upvotes, time posted, etc.). Replies may be displayed “loose” (unattached to their parents), in tree form (attached to their parents), or hidden (only root comments are shown). Editors’ Picks may be pinned to the top of the discussion or unpinned, and replies to pinned comments are hidden for presentation purposes. By default *Der Standard* displays com-

ments in reverse-chronological order, with Editors’ Picks pinned and replies displayed in tree form.

We evaluate fourteen substantive primary orderings: relative votes (upvotes - downvotes), upvotes, chronological, reverse chronological, and ten predictive orderings. Predictive comment-ranking models are carried forward from (Anonymous 2026): conditional logit, XGBoost with and without BGE-M3 text features, and neural ranking models with and without BGE-M3 text features. Each model variant supplies audience- and editor-score orderings: the editorial labels are observed Editors’ Picks, while the audience labels select an equally sized top- k set by relative votes. The models share 41 scalar textual, contextual, author-history, reply-structure, and deliberative predictors (21 non-AQuA inputs plus 20 raw expected AQuA dimension scores; Appendix A). Text-enabled variants additionally use fixed BGE-M3 comment embeddings (Chen et al. 2024). Conditional logit uses article-selector strata; boosted trees and neural networks learn ranking scores. We use the predicted comment scores from the models for ordering, without refitting to the held-out data. An explicit random ordering supplies a fifteenth condition. Crossing these with three reply modes and two pinning states gives $15 \times 3 \times 2 = 90$ final algorithms: 84 substantive algorithms and six random controls.

For FORUM, substantive ordering scores can tie for several comments, so we calculate ten seeded deterministic tie-breaking draws and average them. The random controls have no score ties; we instead generate 100 seeded permutations. These repeated presentations average over arbitrary ordering choices, while the separate 2,000 discussion-bootstrap draws quantify uncertainty across discussions.

Outcome features We evaluate ten primary comment-level outcomes (Table 1), each scored separately rather than collapsed into a single quality index. Together they span deliberative quality and safety, topical relevance and novelty, affect, language form and readability, and participant history. This breadth exposes normative trade-offs—for example, a policy may increase comment–article relevance while reducing novelty—without treating any one measurable attribute as the definition of a good comment.

Measuring feature gatekeeping

FORUM The subsequent analysis of different ranking algorithms is partly inspired by Young et al. (2021), who compared sorting policies on a debating website with regard to winning arguments in argumentation theory. For each discussion, we order the comments according to each ranking policy. We use the “Feature-Oriented Ranking Utility Metric” (FORUM) that measures how well a ranking policy p surfaces the highest-scoring comments in a discussion according to a feature f . The metric compares how the ordering of comments o^p under policy p compares to the best-possible ordering (o^b - descending order of score f), worst-possible ordering (o^w - ascending order of score f), and the expected random baseline. Here we describe how FORUM is calculated for a ranking policy p with feature f on a particular discussion on a news article.

¹“Editors” here includes the journalist(s) who wrote the article. “Community management” are *Der Standard* employees in charge of a small number of volunteer “Community guides” who also assist with content moderation (Der Standard 2021)

Outcome	Operationalisation
AQuA deliberative quality	Weighted composite of 20 expected ordinal AQuA dimension scores (each scored 0–3), on a 0–5 scale (Behrendt et al. 2024).
Comment–article similarity	Mean of the three largest cosine similarities between the comment’s BGE-M3 embedding and embeddings of article passages (nonempty title, subtitle, or body-paragraph segments).
Toxicity	Maximum predicted toxicity probability across comment chunks from the TextDetox XLM-R Large Toxicity Classifier v2 (TextDetox 2025).
Author incumbency	$\log(1 + n_{30})$, where n_{30} is the author’s number of comments on other articles in the preceding 30 days.
Prior audience reception	$R_{30} = \log((U_{30} + 0.5)/(D_{30} + 0.5))$, where U_{30} and D_{30} are the number of up-/downvotes on the author’s comments on other articles in the preceding 30 days.
Comment novelty	Mean cosine distance to the five nearest other comments in the completed discussion, excluding the comment itself.
Positive and negative sentiment	Separate positive and negative probabilities from CardiffNLP XLM-R; long comments use a token-weighted mean over chunks (Barbieri, Espinosa Anke, and Camacho-Collados 2022).
Lexical diversity	Corrected type–token ratio, CTTR = $V/\sqrt{2N}$, where V is the number of distinct word types and N the number of word tokens (Benoit et al. 2018).
Reading difficulty	Raw SMOG-DE, $\sqrt{30P/S} - 2$, where P is the number of polysyllabic words and S the number of sentences, with German syllabification (Benoit et al. 2018; Bamberger and Vanacek 1984).

Table 1: Ten primary comment-level outcomes. The sentiment row contains two separately analysed outcomes.

Let t_i^p denote the cumulative feature value of f through to comment at rank i under policy p , and let T be the total over N candidates. The expected random trajectory is $t_i^r = iT/N$. Descending and ascending feature orders give the attainable upper and lower trajectories t_i^b and t_i^w . The normalised cumulative score at rank i is

$$\gamma_i^p = \begin{cases} (t_i^p - t_i^r)/(t_i^b - t_i^r), & t_i^p > t_i^r, \\ (t_i^p - t_i^r)/(t_i^r - t_i^w), & t_i^p < t_i^r, \\ 0, & t_i^p = t_i^r. \end{cases} \quad (1)$$

This is visualised in Figure 1. If a policy tends to return higher scored comments earlier than expected at random, the line tracking cumulative score will fall above the diagonal baseline. If the policy tends to return higher scoring comments later than expected at random, then it will fall below the baseline. The normalised deviation γ_i^p compares the cumulative feature value through rank i with the best-, worst-, and random-order trajectories. It therefore measures the cumulative prominence of the first i comments, rather

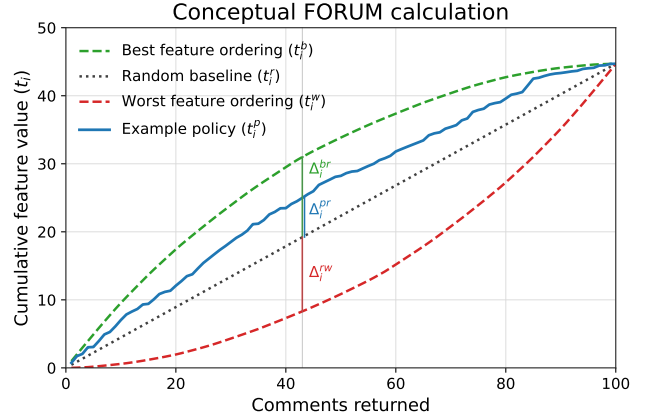


Figure 1: Conceptual FORUM calculation. $\Delta_i^{xy} = t_i^x - t_i^y$. The cumulative feature value under an example policy is compared with the random baseline and the attainable best and worst feature orderings; the normalised deviation at rank i is then averaged through depth d .

than how the i comments themselves are ordered. We measure how a policy p ranks up to depth d by taking the *average* γ_i^p over all ranks through d . Thus, FORUM at depth d is:

$$F_d^p = \frac{1}{d} \sum_{i=1}^d \gamma_i^p. \quad (2)$$

In our analysis, we measure FORUM at two depths: the first ten comments and the full discussion. FORUM lies in $[-1, 1]$: values near 1 indicate an ordering close to the best attainable feature trajectory, values near -1 one close to the worst, and values near zero no systematic departure from random. Because it uses feature values and separate best- and worst-side normalisation, it is not simply a rank-correlation measure. Whether a positive score is desirable depends on the outcome (e.g. with toxicity). Because the random, best, and worst trajectories coincide at N , the final rank is degenerate; we therefore define the full-discussion score as the average through the last nontrivial rank, F_{N-1}^p .

In cases where a ranking policy removes comments from immediate view, such as the “hide replies” functionality, we choose to interpolate a straight line from the cumulative feature score of the final visible comment up to the total cumulative feature score T in the process of calculating F_d . This allows comparison between ranking policies over a different number and set of comments. Practically, this is akin to a situation where any one of the hidden comments is equally likely to be viewed next by a reader (if they choose to click any to expand). Positive and negative deviations have different attainable denominators, so realised random permutations need not average to exactly zero. We therefore retain empirical random controls when interpreting policy contrasts.

The top-10 / full discussion FORUM scores use discussion-averaged scores for each ranking policy and outcome feature. To understand the systematic effect of each

factor in a ranking algorithm, we use paired contrasts to estimate the average effect of each. Ordering contrasts compare each ordering with random while averaging over the six reply/pinning variants. Reply contrasts compare trees or hidden replies with loose presentation, averaging over ordering and pinning. Pinning contrasts average over ordering and reply mode. We retain detailed effects for all policy combinations in Appendix B. We use 2,000 seeded discussion-bootstrap draws for percentile intervals.

Clustering policies by presented feature profiles To gain a greater understanding of how different ranking algorithms are similar / dissimilar from each other, we perform a clustering analysis of them based on the FORUM scores for all 41 of the common scalar features for the predictive models (as opposed to the 10 outcome features previously studied). Each of the 90 algorithms is represented by its discussion-averaged FORUM scores on each of the 41 feature axes (separate clusterings performed for top-10 and full discussion). At each depth, a ten-dimensional UMAP embedding (McInnes et al. 2018) with 15 neighbours and minimum distance zero is clustered with HDBSCAN (Campello, Moulavi, and Sander 2013), using minimum cluster size six, minimum samples three, and excess-of-mass cluster selection. We visualise the resulting clusters in two-dimensional UMAP embeddings (15 neighbours, minimum distance 0.15).

Measuring the presented topic agenda

Article-comment topic model We fit BERTopic (Groendorst 2022) to BGE-M3 embeddings (Chen et al. 2024) of 48,887 article passages (a nonempty article title, subtitle, or body-paragraph segment) and an equally sized sample of 48,887 published comments from the held-out corpus. Comments are sampled round-robin across discussions in hashed order: we take the first hashed comment from each discussion, then the second, and so on, with stable hash tie-breaking within each round. This limits the influence of large discussions and makes the sample reproducible from the same corpus and hashing procedure. The mixed corpus gives article passages and sampled comments equal weight in topic discovery.

Appendix C reports the coverage/stability comparison across different parameters used to assess the configuration. The selected configuration uses cosine-distance UMAP with ten neighbours and five components, followed by HDBSCAN with minimum cluster size 15, minimum samples five, and seed 2026. Automatic topic reduction yields 439 non-outlier topics; outliers remain unassigned. The final transformed corpus contains 48,887 article passages and 1,203,611 comments, of which 53.59% and 26.64%, respectively, receive valid topic assignments. The configuration balances stability, coverage, and topic granularity (Appendix C). The low comment coverage can be largely explained through a large number of true outlier non-substantive comments, e.g., “I agree”, “What’s your point?”. The agenda analysis includes 2,531 discussions with both an article and a complete-discussion topic distribution. This is a smaller sample than the feature analysis because both distri-

butions require at least one topic assigned document. Topic-distribution mass for an article / discussion is renormalised to exclude topic outliers.

Vote-activity weighting We estimate a proxy for attention at each rank from total votes (upvotes plus downvotes) at each visible rank under reconstructed reverse-chronological/tree/pinned presentation (default at *Der Standard*). Within each development discussion, each visible rank receives its share of that discussion’s visible vote total; we then average these shares equally across discussions and fit a smoothed empirical spline on 2,553 discussions. Hidden comments contribute neither ranks nor votes. The length-conditioned model is

$$p(r | L) = \frac{w_r}{\sum_{u=1}^L w_u}, \quad (3)$$

where r is visible rank, L is the number of comments visible under policy p , and w_r is the frozen nonnegative rank weight. Log weights are represented by a cubic B-spline of log rank with a second-difference penalty. Nested development-fold validation selects smoothing; the final spline is refitted on all development discussions. For each policy, we renormalise the frozen weights over its visible length L , so they describe relative allocation among the comments exposed by that policy; omitted comments receive no visible rank or weight. Appendix D gives the fit and its length-adjusted curves. These weights are a simple estimate of attention, not measured attention or a causal position effect. Votes combine age, content, prior prominence, and social feedback, among other factors which we are unable to capture or account for.

We compute article and complete-discussion topic distributions (unaffected by ranking). A_j is the equally weighted mean of valid article-passage topic vectors for discussion j , and C_j is the equally weighted mean of valid topic vectors over all eligible comments in that discussion; outliers are excluded and each distribution is renormalised. For a given ranking algorithm, we employ a weighted average of the topics of the visible comments, with weights reflecting expected attention at each rank. For discussion j , policy p , and presentation draw b , let q_{jprb} be the topic vector of the comment at visible rank r , w_r be the weight for that rank, and a_{jprb} indicate a valid topic assignment. The visible topic distribution is

$$V_{jpb} = \frac{\sum_{r=1}^{L_{jpb}} w_r a_{jprb} q_{jprb}}{\sum_{r=1}^{L_{jpb}} w_r a_{jprb}}, \quad (4)$$

where L_{jpb} is the number of visible comments. Outlier comments may occupy ranks and consume weight but do not contribute topic mass. We retain ten pseudo-random permutations generated from fixed seeds, making substantive tie-breaking reproducible; random controls use 25 seeded permutations. Paired 95% percentile intervals use 2,000 discussion-bootstrap draws. This visible topic distribution captures the visible weighted topic agenda under a ranking policy, as opposed to the full discussion unweighted topic distribution.

Article versus relative-vote agenda similarity We measure how each policy changes similarity to two reference agendas: the article distribution T_j^{article} and a relative-vote distribution T_j^{votes} , obtained by averaging the weighted visible distributions over ten relative-vote draws with loose replies and no pinning. Let R denote random ordering with loose replies and no pinning. For either target T_j , the cosine gain is

$$g_{jp}(T_j) = \mathbb{E}_b \cos(T_j, V_{jpb}) - \mathbb{E}_b \cos(T_j, V_{jRb}). \quad (5)$$

Positive gain indicates greater similarity to the target than under the common random baseline. Article alignment is measured by $g_{jp}(T_j^{\text{article}})$; the difference $g_{jp}(T_j^{\text{article}}) - g_{jp}(T_j^{\text{votes}})$ indicates whether a policy gains more towards the article or the vote-derived agenda.

For both outcomes, we report average paired effects across discussions. Ordering effects compare each ordering with random within the same reply/pinning configuration, then average over the six configurations. Reply effects compare trees or hidden replies with loose presentation; pinning effects compare pinned with unpinned presentation, averaging over substantive orderings and the remaining factor. The oracle is evaluated on both outcomes against R . Intervals use 2,000 discussion-bootstrap draws. The two-target scatter instead shows mean gains for each complete policy bundle; points below its diagonal have larger article-directed gains.

Article agenda oracle A discussion’s available comments may not permit an ordering that matches or even gets close to the article’s topic distribution. We therefore construct an article-aware, best-found benchmark under the same rank-weighting scheme. The heuristic tries to maximise article-visible cosine similarity by ordering comments without any constraints. Greedy refinement starts from the article-affinity order, the 14 structured policy orders, and eight seeded random permutations, with up to five refinement iterations, four perturbation attempts per iteration, a 10% perturbation fraction, and two local moves. Invalid-topic comments may occupy positions but contribute no topic mass. This result is a heuristic benchmark for comparison rather than a mathematical upper bound on attainable alignment.

Results

Algorithmic gatekeeping

The feature analysis addresses RQ1 by comparing ordering and interface effects at two depths.

Ordering and interface effects on presented features

We highlight AQuA, article similarity, and CTTR as they represent deliberative quality, article alignment, and textual form; semantic novelty illustrates a countervailing effect. Figure 2 shows average ordering and interface effects across all ten outcomes. At top 10, regression-editor ordering raises the FORUM score for AQuA by 0.2691 (95% CI 0.2652–0.2727), article similarity by 0.2115 (0.2077–0.2154), and CTTR by 0.2605 (0.2568–0.2643). Relative-vote ordering produces smaller gains for the same outcomes: 0.0855 (0.0810–0.0897), 0.0741 (0.0701–0.0780), and 0.0794 (0.0755–0.0835).

Interface effects are likewise feature-specific. Relative to loose presentation at top 10, intact reply trees reduce the FORUM score for AQuA by 0.1219 (95% CI −0.1251 to −0.1187), article similarity by 0.1268 (−0.1295 to −0.1239), and CTTR by 0.1181 (−0.1207 to −0.1153), while increasing semantic novelty by 0.0931 (0.0900–0.0958). Over the full trajectory, the tree penalty for article similarity is larger (−0.2002, −0.2036 to −0.1968), whereas hiding replies reduces AQuA by 0.0402 (−0.0420 to −0.0384) and pinning reduces AQuA by 0.0197 (−0.0216 to −0.0177). Thus the same policy can look different at different exposure horizons. Positive FORUM means earlier prominence of larger values of the named feature; it does not mean that every larger value is desirable, as with toxicity.

Across the ten outcomes, predictive models consistently increase deliberative quality, article similarity, lexical diversity, reading difficulty, and prior audience reception at both depths, while reducing semantic novelty. Sentiment, toxicity, and author incumbency show weaker and more model- and horizon-dependent patterns. The three highlighted outcomes therefore illustrate the common positive movement, while novelty and the remaining outcomes show why it cannot be interpreted as a general improvement.

The default presentation The default at *Der Standard* combines reverse-chronological ordering, reply trees, and pinned Editors’ Picks. Its top-10 FORUM scores indicate earlier prominence for AQuA (0.303), article similarity (0.293), and CTTR (0.268), alongside suppression of toxicity (−0.385) and semantic novelty (−0.226). These patterns weaken over the full discussion: the corresponding scores are 0.078, 0.114, 0.089, 0.043, and −0.042, with toxicity changing sign. Prior audience reception also changes sign, from 0.034 to −0.023.

Policies with similar overall effects Clustering the 41-feature FORUM profiles yields four top-10 clusters, containing 32, 18, 22, and 18 distinct ranking algorithms, with no outliers (Fig. 3). At full depth, the same procedure identifies seven clusters and four noise points. Across both depths, pin status (fill) and reply mode (shape) appear to act as strong separators in this space, with smaller scale groupings by ordering (colour). At top 10, one cluster contains all 15 unpinned tree bundles and another contains all 15 pinned tree bundles; the learned loose and hidden bundles are further separated largely by pinning, with 18 of the 20 unpinned learned loose/hidden bundles forming a distinct cluster. In the full depth clustering, one cluster contains all twenty loose-presentation algorithms based on predictive models. Hidden-reply bundles occupy distinct groups. The four outlier points are the loose relative-vote and upvote policies with and without pinning.

Algorithmic agenda-setting

This analysis addresses RQ2 by comparing the alignment of the presented topic distributions.

Topic agenda shifts Figure 4 shows average policy effects on article-directed cosine gain. Predictive mod-

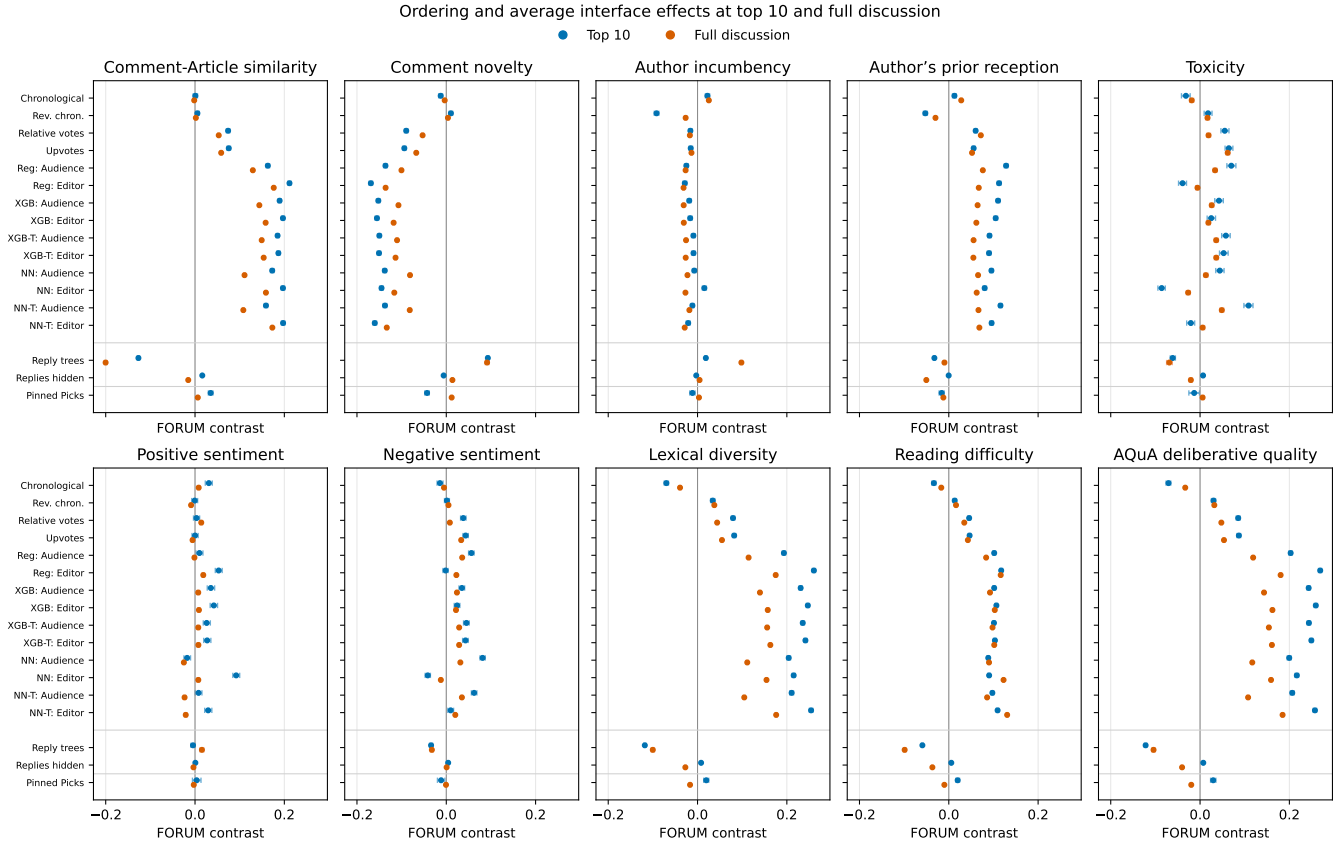


Figure 2: Ordering and average interface effects for ten primary attributes at both depths. Points compare orderings with random; interface contrasts average across the other policy factors. Points and intervals are paired 95% discussion-bootstrap intervals.

els increase alignment relative to random ordering under matched reply/pinning configurations. The regression-editor ordering has an effect of $+0.0077$ (95% CI 0.0063 – 0.0091), and the neural-editor ordering has an effect of $+0.0075$ (0.0061 – 0.0089). Relative votes and upvotes produce smaller point estimates, $+0.0019$ (0.0007 – 0.0030) and $+0.0016$ (0.0005 – 0.0028). Chronological and reverse-chronological ordering have intervals spanning zero. For scale, the mean random/loose/unpinned-to-oracle gap is 0.0860 ; the conditional-logit and neural editor effects are about 8.9% and 8.7% of that benchmark gap.

Reply handling produces a larger shift. Hiding replies increases article-directed cosine gain by $+0.0286$ (0.0252 – 0.0318), whereas intact trees reduce it by -0.0036 (-0.0043 to -0.0030). Pinning increases it by $+0.0027$ (0.0011 – 0.0043). Relative to the same 0.0860 benchmark gap, the hidden-reply and pinning effects are about 33.2% and 3.1%; intact trees move in the opposite direction. These ratios compare effect magnitudes with a common heuristic benchmark; they are not fractions of a known optimum attained by a particular policy bundle.

Article versus relative-vote agenda gains The two-target comparison shows that article-directed movement is not simply movement toward any salient agenda (Figures 4 and

5). For every predictive model, the marginal effect on article gain exceeds its effect on relative-vote gain, with all ten paired difference intervals above zero. The regression-editor difference is $+0.0076$ (95% CI 0.0060 – 0.0092); corresponding differences are $+0.0064$ for XGBoost editor ordering (0.0047 – 0.0079) and $+0.0075$ for neural editor ordering (0.0058 – 0.0092). The text-enabled editor differences remain positive, $+0.0045$ for XGBoost (0.0030 – 0.0060) and $+0.0056$ for the neural model (0.0040 – 0.0073). The article-optimising oracle gains 0.0860 towards the article and -0.0241 towards the vote-derived agenda, giving an article-minus-vote benchmark of 0.1100 in the combined figure. This ordering was optimised for article alignment, not for the gain difference.

This pattern has clear limits. The relative-vote and up-vote ordering contrasts favour the vote-derived target (as one would expect), with article-minus-vote differences of -0.0018 (-0.0031 to -0.0005) and -0.0020 (-0.0034 to -0.0008). Chronological ordering also has a negative difference, while reverse chronology is inconclusive. Hiding replies strongly favours the article target ($+0.0474$, 0.0429 – 0.0518), but retaining discussion trees favours the vote-derived target (-0.0028 , -0.0036 to -0.0021). Pinning’s two-target difference spans zero. Thus the evidence supports stronger article-directed effects for predictive model

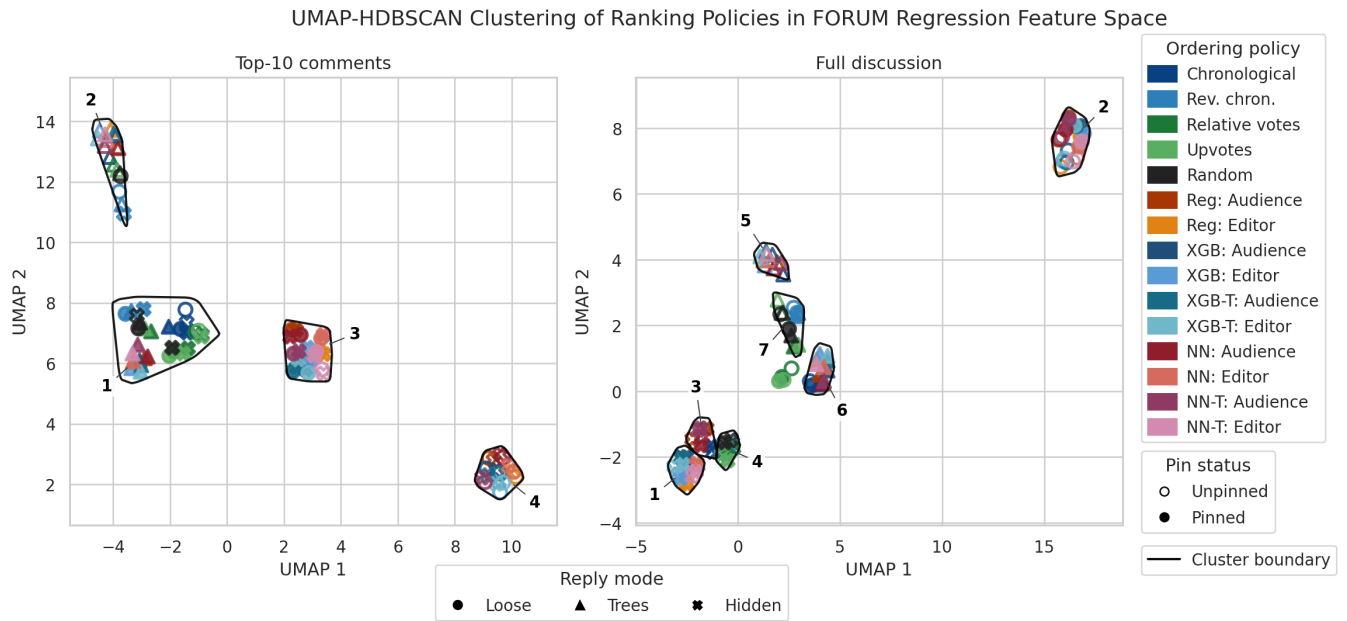


Figure 3: UMAP-HDBSCAN clustering of policies in FORUM regression-feature space, separately at top 10 and full depth. Clusters are fitted in ten-dimensional UMAP space and displayed in a separate two-dimensional UMAP embedding.

and hidden-reply presentation, rather than for every interface decision.

For the complete default policy, mean article gain against random/loose/unpinned presentation is 0.0118 (95% CI 0.0097–0.0140), compared with 0.0023 (0.0015–0.0031) towards the relative-vote agenda. The paired article-minus-vote gain is 0.0095 (0.0074–0.0117), across 2,530 discussions with valid gains for both targets. Thus the default moves towards both reference agendas, with a larger gain towards the article. These bundle-level gains correspond to a point in Figure 5, rather than a marginal effect in Figure 4.

Discussion

Ranking policy is an editorial-governance decision about the representation of an existing comment pool. Journalists exert curatorial gatekeeping through the comments they pin, while ordering and reply handling shape the surrounding discussion. Our results show how these choices alter both the attributes brought forward (RQ1) and the topics receiving prominence (RQ2).

For RQ1, predictive models promote AQuA, article similarity, and lexical diversity; interface choices can offset these gains. These outcomes overlap with ranker inputs, so gains do not independently validate quality. Because only 29.7% of candidates are roots, reply handling is consequential: trees preserve context but constrain early prominence, whereas hiding replies increases article alignment while reducing full-discussion AQuA. These trade-offs, and differences between top-10 and full-discussion results, show why ranking audits must specify both interface and evaluation depth.

Pinning gives editors targeted control over early prominence. Editors’ Picks account for only 0.87% of comments,

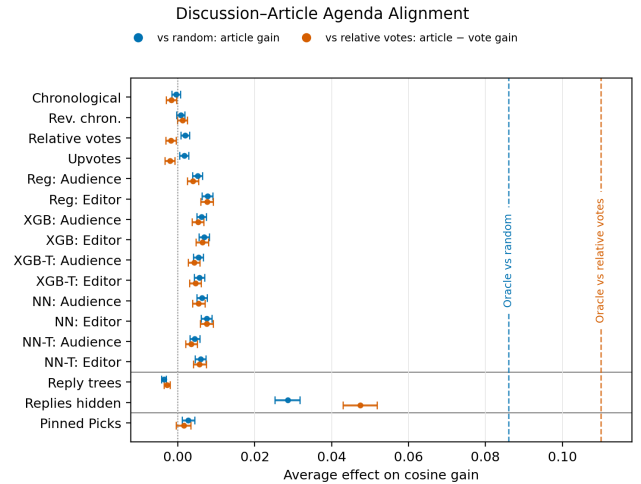


Figure 4: Average policy effects on article gain versus random and article-minus-relative-vote gain. Both series use paired ordering and interface contrasts; points and intervals are 95% discussion-bootstrap intervals.

with an average of four per discussion, yet occupy the first positions when pinned. Their feature effects are small in comparison to full discussion ordering or reply handling, yet one must bear in mind that this is an intervention over a very small number of comments. Pinning effects can change direction over the full discussion: the AQuA FORUM contrast is positive at top 10 (0.0294, 95% CI 0.0235–0.0355) but negative at full depth (−0.0197, −0.0216 to −0.0177). Future research could examine if larger curated selections

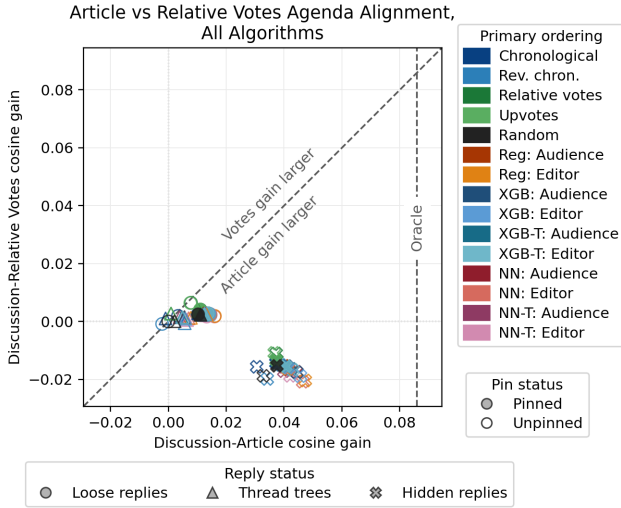


Figure 5: Mean cosine gains toward the article and relative-vote topic agenda for all 90 ranking algorithms.

change this relationship.

Greater article alignment can come with lower deliberative quality: hiding replies increases article alignment while reducing full-discussion AQUA. Further work should examine how these trade-offs affect conversation interpretability and reader experience, including the value of seeing a contribution alongside the comment it answers.

For RQ2, the topic analysis shows that predictive models yield larger article-directed than relative-vote-directed gains in the paired ordering contrasts. Hiding replies strengthens article alignment further and shifts it away from the vote-derived agenda. A newsroom can therefore obtain a more article-focused presentation at the expense of breadth and the prominence of issues introduced or endorsed by commenters. Pinning does not show the same consistent conclusive preference between the two targets.

Neither target provides a neutral account of public opinion. The article distribution reflects the issues covered in the story, while the relative-vote distribution reflects participating voters under existing exposure conditions. Topical agreement also does not imply agreement in viewpoint. A newsroom seeking rapid orientation may choose differently from one seeking to preserve extended disagreement and exchange. We advise against optimising solely for one FORUM score or agenda. The sizeable effects for predictive models show their capacity to reshape a discussion, but whether a change improves participation or readers’ understanding requires behavioural evidence. These questions also apply to news discussion on social media, where ranking may play an even wider role in allocating visibility.

Changes in topic prominence and visible comment attributes correspond to potential first- and second-level agenda-setting mechanisms, respectively; because we measure presentation rather than audience response, their behavioural effects remain to be established.

Our versatile FORUM score can be applied in future re-

search, including different news platforms and experimental studies on ranking algorithms’ impact on user behaviour. It can also be used to study ranking in other contexts where a finite set of items, each with measurable features, is ordered by a potentially unknown algorithm. Examples include news feeds, social media comments, job postings, and product listings. In each case, the choice of features and the treatment of omitted items must reflect the setting being studied.

Limitations and Ethical Considerations

We measure presentation rather than behavioural response on one German-language forum, limited to discussions containing Editors’ Picks. Generalisation and causal effects require studies on other platforms and experimental or policy-change designs. Our snapshot also omits changes over time in votes, pinning, and deletions.

Topic assignments depend on the embedding model, mixed fitting sample, and clustering choices. Only 26.64% of comments receive a valid assignment, so the topic results describe the assigned portion of discussion. The relatively high proportion of unassigned comments is unsurprising given the large number of short, non-substantive comments in the corpus. Yet, these comments may contain implicit endorsement of specific viewpoints or agenda issues (e.g. “We should be talking more about this!”). The topic model is also not a measure of political diversity or agreement.

The study received institutional ethics review approval (ID 13838441). The project follows the Association of Internet Researchers’ ethics framework (franzke et al. 2020) and, whilst it is not feasible or methodologically prudent to seek informed consent at scale, considers contextual privacy and potential harm throughout the research process. Data collection followed the site’s robots.txt directives and terms (Der Standard 2026a,b). The collector retrieved public records without posting comments, voting, or otherwise participating in discussions. Author identifiers were replaced by hashed pseudonyms during collection. This enables longitudinal linkage but does not anonymise the data: verbatim text, timestamps, and histories may remain identifying or searchable. We share code, documentation, and aggregate results while withholding the full comment-level dataset to reduce risks to commenters’ privacy.

Ranking audits can support more transparent moderation, but the same methods could be used to suppress viewpoints or engineer apparent consensus. Their appropriate use requires explicit objectives and disclosure of measurement limits. Newsrooms should make the purpose of ranking legible to readers and consider whose contributions lose prominence. Article alignment and automated quality scores should not be treated as grounds for excluding legitimate criticism; preserving routes to wider discussion can help readers assess comments for themselves.

Conclusion

This paper examines the representational consequences of comment ranking on a large news website. FORUM measures how policies promote or suppress comment attributes,

while topic modelling assesses changes in the topical focus of the presented discussion. For RQ1, predictive models bring forward deliberative quality, article similarity, and lexical diversity, but reply handling can offset these gains, and early prominence differs from full-discussion presentation. For RQ2, predictive models show stronger article-directed than vote-directed marginal gains, while hiding replies further intensifies article alignment.

Journalists retain direct editorial influence through pinned Editors' Picks, but ordering and reply handling also shape which contributions receive prominence. No single policy is best across the measured attributes and agendas. Newsrooms must choose in light of their purposes for discussion, while researchers and readers need to account for display rules when interpreting visible responses to news.

News comments allow readers to contribute knowledge, experience, and criticism to journalism. Making the consequences of ranking explicit can help newsrooms support that participation while attending to contributions that their chosen presentation leaves less visible.

Acknowledgments

OpenAI Codex was used in co-writing analysis code and re-organising author-written manuscript text.

References

- Althuniya, N.; Liu, X.; Sirrianni, J. W.; and Adams, D. 2020. Opinion Discovery Framework: Toward a Quality Opinion-Centric Platform. *Journal of Advances in Information Technology*, 11(2).
- Anonymous. 2026. The News Comment Gap: Editorial and Audience Preferences in Online News Discussions. Companion paper, under review at ICWSM.
- Atwater, T.; Fico, F.; and Pizante, G. 1987. Reporting on the state legislature: A case study of inter-media agenda-setting. *Newspaper Research Journal*, 8(2): 53–61.
- Bamberger, R.; and Vanacek, E. 1984. *Lesen-Verstehen-Lernen-Schreiben*. Diesterweg.
- Bandy, J.; and Diakopoulos, N. 2020. Auditing News Curation Systems: A Case Study Examining Algorithmic and Editorial Logic in Apple News. *Proceedings of the International AAAI Conference on Web and Social Media*, 14(1): 36–47.
- Barbieri, F.; Espinosa Anke, L.; and Camacho-Collados, J. 2022. XLM-T: Multilingual Language Models in Twitter for Sentiment Analysis and Beyond. In Calzolari, N.; Béchet, F.; Blache, P.; Choukri, K.; Cieri, C.; Declerck, T.; Goggi, S.; Isahara, H.; Maegaard, B.; Mariani, J.; Mazo, H.; Odijk, J.; and Piperidis, S., eds., *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, 258–266. Marseille, France: European Language Resources Association.
- Behrendt, M.; Wagner, S. S.; Ziegele, M.; Wilms, L.; Stoll, A.; Heinbach, D.; and Harmeling, S. 2024. AQuA – Combining Experts' and Non-Experts' Views To Assess Deliberation Quality in Online Discussions Using LLMs. In Hautli-Janisz, A.; Lapesa, G.; Anastasiou, L.; Gold, V.; Liddo, A. D.; and Reed, C., eds., *Proceedings of the First Workshop on Language-driven Deliberation Technology (DELITE) @ LREC-COLING 2024*, 1–12. Torino, Italia: ELRA and ICCL.
- Benoit, K.; Watanabe, K.; Wang, H.; Nulty, P.; Obeng, A.; Müller, S.; and Matsuo, A. 2018. quantda: An R package for the quantitative analysis of textual data. *Journal of Open Source Software*, 3(30): 774–774.
- Bruns, A. 2003. Gatewatching, Not Gatekeeping: Collaborative Online News. *Media International Australia*, 107: 31–44.
- Burger, C. 2022. Von zehn auf 20 Millionen in zwei Jahren. <https://www.derstandard.de/story/2000132276894/von-zehn-auf-20-millionen-in-zwei-jahren>. Accessed 14-September-2026.
- Campello, R. J. G. B.; Moulavi, D.; and Sander, J. 2013. Density-Based Clustering Based on Hierarchical Density Estimates. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining*, 160–172. Springer.
- Chan, J.; Choi, F.; Saha, K.; and Chandrasekharan, E. 2026. Examining Algorithmic Curation on Social Media: An Empirical Audit of Reddit's r/popular Feed. *Proceedings of the International AAAI Conference on Web and Social Media*, 20(1): 391–406.
- Chen, J.; Xiao, S.; Zhang, P.; Luo, K.; Lian, D.; and Liu, Z. 2024. M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. In Ku, L.-W.; Martins, A.; and Srikumar, V., eds., *Findings of the Association for Computational Linguistics: ACL 2024*, 2318–2335. Bangkok, Thailand: Association for Computational Linguistics.
- Cheng, J.; Danescu-Niculescu-Mizil, C.; and Leskovec, J. 2014. How community feedback shapes user behavior. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 8, 41–50.
- Der Standard. 2021. Community-Guides. <https://www.derstandard.at/story/2000125444821/community-guides-aus-der-community-fuer-die-community>. Accessed 14-September-2026.
- Der Standard. 2026a. Allgemeine Geschäftsbedingungen - Text- und Data-Mining, Crawler. <https://about.derstandard.at/agb/#Text-DataMining-Crawler>. Accessed 14-September-2026.
- Der Standard. 2026b. robots.txt. <https://www.derstandard.at/robots.txt>. Accessed 16-September-2026.
- Diakopoulos, N. 2015. Picking the NYT picks: Editorial criteria and automation in the curation of online news comments. *ISOJ Journal*, 6(1): 147–166.
- Diakopoulos, N.; and Naaman, M. 2011. Towards quality discourse in online news comments. In *Proceedings of the ACM 2011 conference on Computer supported cooperative work*, 133–142.
- Efstratiou, A.; Duskin, K.; Starbird, K.; and Spiro, E. S. 2026. Rabble-Rousers in the New King's Court: Algorithmic Effects on Account Visibility in Pre-X Twitter. *Proceedings of the International AAAI Conference on Web and Social Media*, 20(1): 691–705.

- Einarsson, A. M.; Helles, R.; and Lomborg, S. 2025. Algorithmic agenda-setting: the subtle effects of news recommender systems on political agendas in the Danish 2022 general election. *Information, Communication & Society*, 28(2): 218–238.
- franzke, a. s.; Bechmann, A.; Ess, C. M.; and Zimmer, M. 2020. Internet research: Ethical guidelines 3.0. Technical report.
- Fraxanet, E.; Gómez, V.; Kaltenbrunner, A.; and Pellert, M. 2026. A Decade of News Forum Interactions: Threaded Conversations, Signed Votes, and Topical Tags. *Scientific Data*, 13(1): 1193.
- Gebru, T.; Morgenstern, J.; Vecchione, B.; Vaughan, J. W.; Wallach, H.; Daumé III, H.; and Crawford, K. 2021. Datasheets for Datasets. *Communications of the ACM*, 64(12): 86–92.
- Grootendorst, M. 2022. BERTopic: Neural Topic Modeling with a Class-Based TF-IDF Procedure. *arXiv preprint arXiv:2203.05794*.
- Guo, L.; and McCombs, M. 2011. Network agenda setting: A third level of media effects. In *annual conference of the International Communication Association, Boston, MA*.
- He, Y.; Gu, Y.; Shekhar, R.; Castro, I.; and Tyson, G. 2024. Making the Pick: Understanding Professional Editor Comment Curation in Online News. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 18, 557–568.
- Juarez Miro, C. 2022. The comment gap: Affective publics and gatekeeping in The New York Times’ comment sections. *Journalism*, 23(4): 858–874.
- Kim, S.; and Noh, S. 2026. Disproportionate Voices: Participation Inequality and Hostile Engagement in News Comments. *Proceedings of the International AAAI Conference on Web and Social Media*, 20(1): 1256–1272.
- Kim, S.-T.; and Lee, Y.-H. 2006. New functions of Internet mediated agenda-setting: Agenda-rippling and reversed agenda-setting. *Korean journal of journalism & communication studies*, 50(3): 175–205.
- Lampe, C.; Zube, P.; Lee, J.; Park, C. H.; and Johnston, E. 2014. Crowdsourcing civility: A natural experiment examining the effects of distributed moderation in online forums. *Government Information Quarterly*, 31(2): 317–326.
- Levy, D. M. 2008. *Information overload*, 497–513. Wiley.
- Masullo Chen, G.; and Pain, P. 2017. Normalizing online comments. *Journalism Practice*, 11(7): 876–892.
- McCombs, M.; Llamas, J. P.; Lopez-Escobar, E.; and Rey, F. 1997. Candidate images in Spanish elections: Second-level agenda-setting effects. *Journalism & mass communication quarterly*, 74(4): 703–717.
- McCombs, M.; and Shaw, D. L. 1972. The agenda-setting function of mass media. *Public opinion quarterly*, 36(2): 176–187.
- McInnes, L.; Healy, J.; Saul, N.; and Grossberger, L. 2018. UMAP: Uniform Manifold Approximation and Projection. *Journal of Open Source Software*, 3(29): 861.
- Momeni, E.; Cardie, C.; and Diakopoulos, N. 2015. A Survey on Assessment and Ranking Methodologies for User-Generated Content on the Web. *ACM Comput. Surv.*, 48(3).
- Nielsen, R. K. 2016. News media, search engines and social networking sites as varieties of online gatekeepers. In *Rethinking journalism again*, 93–108. Routledge.
- Nutakki, B.; Kempermann, M. L.; and Weber, I. 2026. You See It, They Don’t: An Exploratory Study of User-to-User Variation in Instagram Comments. *Proceedings of the International AAAI Conference on Web and Social Media*, 20(1): 3024–3030.
- Piccardi, T.; Saveski, M.; Jia, C.; Hancock, J.; Tsai, J. L.; and Bernstein, M. S. 2025. Reranking partisan animosity in algorithmic social media feeds alters affective polarization. *Science*, 390(6776): eadu5584.
- Shaw, D. L.; McCombs, M.; Weaver, D. H.; and Hamm, B. J. 1999. Individuals, groups, and agenda melding: A theory of social dissonance. *International journal of public opinion research*, 11(1): 2–24.
- Shoemaker, P. J.; and Vos, T. 2009. *Gatekeeping theory*. Routledge.
- TextDetox. 2025. XLM-R Large Toxicity Classifier v2. Hugging Face model card. Revision cde9d07ac4df2af9c02d2461dee068bf04a58728.
- Trielli, D.; and Diakopoulos, N. 2022. Algorithmic agenda-setting: The shape of search media during the 2020 US election. In *International Symposium on Online Journalism*, volume 12, 45–70.
- Van Dijck, J.; and Poell, T. 2013. Understanding social media logic. *Media and communication*, 1(1): 2–14.
- Voinea, D. V. 2025. Reconceptualizing Gatekeeping in the Age of Artificial Intelligence: A Theoretical Exploration of Artificial Intelligence-Driven News Curation and Automated Journalism. *Journalism and Media*, 6(2): 68.
- Wallace, J. 2018. Modelling contemporary gatekeeping: The rise of individuals, algorithms and platforms in digital news dissemination. *Digital Journalism*, 6(3): 274–293.
- Wang, Y.; and Diakopoulos, N. 2022. Highlighting High-quality Content as a Moderation Strategy: The Role of New York Times Picks in Comment Quality and Engagement. *ACM Transactions on Social Computing (TSC)*, 4(4): 1–24.
- Wang, Y.; Wang, L.; Li, Y.; He, D.; and Liu, T.-Y. 2013. A theoretical analysis of NDCG type ranking measures. In *Conference on learning theory*, 25–54. PMLR.
- White, D. M. 1950. The “gate keeper”: A case study in the selection of news. *Journalism quarterly*, 27(4): 383–390.
- Wilkinson, M. D.; Dumontier, M.; Aalbersberg, I. J.; et al. 2016. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3: 160018.
- Young, A. P.; Joglekar, S.; Boschi, G.; and Sastry, N. 2021. Ranking comment sorting policies in online debates. *Argument & Computation*, 12(2): 265–285.
- Ziegele, M.; Naab, T. K.; and Jost, P. 2020. Lonely together? Identifying the determinants of collective corrective action against uncivil comments. *New Media & Society*, 22(5): 731–751.

Paper Checklist

1. For most authors...

- (a) Would answering this research question advance science without violating social contracts, such as violating privacy norms, perpetuating unfair profiling, exacerbating the socio-economic divide, or implying disrespect to societies or cultures? **Yes.** The research value and privacy risks are discussed in the Introduction and Limitations and Ethical Considerations sections.
 - (b) Do your main claims in the abstract and introduction accurately reflect the paper's contributions and scope? **Yes.** They describe ranking-policy effects on feature and topic prominence and do not claim effects on reader beliefs or behaviour.
 - (c) Do you clarify how the proposed methodological approach is appropriate for the claims made? **Yes.** Data and Methods explains our approaches with their interpretive limits.
 - (d) Do you clarify what are possible artifacts in the data used, given population-specific distributions? **Yes.** Data and Methods and Limitations and Ethical Considerations discuss the single German-language platform, selection of discussions containing Editors' Picks, snapshot votes and pinning, confounded vote-activity weights, and incomplete topic-assignment coverage.
 - (e) Did you describe the limitations of your work? **Yes.** Limitations cover the single-platform corpus, topic coverage and validity, voting-activity proxies, counterfactual interface assumptions, and the absence of behavioural measurement.
 - (f) Did you discuss any potential negative societal impacts of your work? **Yes.** Limitations and Ethical Considerations discusses viewpoint suppression and apparent-consensus engineering, and asks newsrooms to consider whose contributions lose prominence.
 - (g) Did you discuss any potential misuse of your work? **Yes.** Limitations and Ethical Considerations explicitly discusses misuse for viewpoint suppression and apparent-consensus engineering.
 - (h) Did you describe steps taken to prevent or mitigate potential negative outcomes of the research, such as data and model documentation, data anonymization, responsible release, access control, and the reproducibility of findings? **Yes.** Limitations and Ethical Considerations describes implemented author pseudonymisation on collection and aggregate reporting, alongside limited data release.
 - (i) Have you read the ethics review guidelines and ensured that your paper conforms to them? **Yes.**
- ### 2. Additionally, if your study involves hypotheses testing...
- (a) Did you clearly state the assumptions underlying all theoretical results? **Yes.** Data and Methods states the assumptions underlying FORUM, the random-order baseline, hidden-comment interpolation, paired policy contrasts, topic assignment, and vote-derived atten-

tion weights. The paper distinguishes representational comparisons from behavioural or causal effects.

- (b) Have you provided justifications for all theoretical results? **Yes.** Data and Methods defines and motivates FORUM and the topic-alignment measures.
 - (c) Did you discuss competing hypotheses or theories that might challenge or complement your theoretical results? **Yes.** Related Work and Discussion distinguish algorithmic gatekeeping from agenda-setting and contrast journalist-directed, audience-vote-derived, chronological, and discussion-structure accounts of comment prominence.
 - (d) Have you considered alternative mechanisms or explanations that might account for the same outcomes observed in your study? **Yes.** The paper discusses vote confounding, prior prominence and social feedback, missing or hidden comments, topic-assignment coverage, and interface semantics as alternative explanations or limitations.
 - (e) Did you address potential biases or limitations in your theoretical framework? **Yes.** Related Work and Limitations and Ethical Considerations explain why article and vote-derived agendas are incomplete normative references, why topic agreement does not imply viewpoint agreement, and why presentation is not equivalent to reader attention or behavioural influence.
 - (f) Have you related your theoretical results to the existing literature in social science? **Yes.** Related Work and Discussion connect the empirical comparisons to agenda-setting, gatekeeping, online commenting, moderation, and audience-influence research.
 - (g) Did you discuss the implications of your theoretical results for policy, practice, or further research in the social science domain? **Yes.** The discussion calls for explicit ranking objectives, interface-semantic reporting, measurement disclosure, and future behavioural validation.
- ### 3. Additionally, if you are including theoretical proofs...
- (a) Did you state the full set of assumptions of all theoretical results? **N/A.**
 - (b) Did you include complete proofs of all theoretical results? **N/A.**
- ### 4. Additionally, if you ran machine learning experiments...
- (a) Did you include the code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL)? **Yes.** The anonymised repository is available at <https://anonymous.4open.science/r/commentgap-15CB>. Full comment-level data is not released due to the privacy risks described in Limitations and Ethical Considerations.
 - (b) Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)? **Yes.** The paper specifies the held-out corpus, topic-model search, selected clustering settings, vote-activity fit, and policy evaluation. Ranker hyperparameters and

model-selection details are reported in the companion paper’s methods and appendix.

- (c) Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)? [Yes. Figures report paired discussion-bootstrap intervals.](#)
 - (d) Did you include the total amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)? [Yes. Appendix F reports the primary hardware and software environment.](#)
 - (e) Do you justify how the proposed evaluation is sufficient and appropriate for the claims made? [Yes. Data and Methods motivates paired policy contrasts and explicit article- and vote-derived references; the paper states that it does not measure reader behaviour or causal effects.](#)
 - (f) Do you discuss what is “the cost” of misclassification and fault (in)tolerance? [Yes. Limitations and Ethical Considerations cautions against treating automated quality scores or article alignment as grounds for excluding legitimate criticism, and discusses the risks of viewpoint suppression and apparent-consensus engineering. The discussion explains that feature gains involve trade-offs rather than a single notion of quality.](#)
5. Additionally, if you are using existing assets (e.g., code, data, models) or curating/releasing new assets, **without compromising anonymity**...
- (a) If your work uses existing assets, did you cite the creators? [Yes. The paper cites BGE-M3, AQuA, the CardiffNLP XLM-R sentiment classifier, the TextDetox XLM-R Large Toxicity Classifier v2, BERTopic, and the reused ranking methods where they are introduced.](#)
 - (b) Did you mention the license of the assets? [Yes. Appendix E documents the upstream licence declarations for BGE-M3, sentiment, AQuA, and toxicity.](#)
 - (c) Did you include any new assets in the supplemental material or as a URL? [No.](#)
 - (d) Did you discuss whether and how consent was obtained from people whose data you’re using/curating? [Yes. Limitations and Ethical Considerations explains the public, non-participatory collection and adherence to the AoIR ethics framework.](#)
 - (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? [Yes. Limitations and Ethical Considerations explains that verbatim text, timestamps, and author histories may remain identifying or searchable and that hashed pseudonyms do not anonymise the data.](#)
 - (f) If you are curating or releasing new datasets, did you discuss how you intend to make your datasets FAIR (see Wilkinson et al. (2016))? [N/A. Not releasing a new dataset.](#)
 - (g) If you are curating or releasing new datasets, did you create a Datasheet for the Dataset (see Gebru et al. (2021))? [N/A.](#)
6. Additionally, if you used crowdsourcing or conducted research with human subjects, **without compromising anonymity**...
- (a) Did you include the full text of instructions given to participants and screenshots? [N/A.](#)
 - (b) Did you describe any potential participant risks, with mentions of Institutional Review Board (IRB) approvals? [N/A.](#)
 - (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? [N/A.](#)
 - (d) Did you discuss how data is stored, shared, and de-identified? [N/A.](#)

A Predictors and Model Development

Table 2 details the common predictor set introduced in the “Constructing ranking algorithms” subsection. Numeric predictors are standardised using development data; binary indicators remain on the 0–1 scale. Predictor novelty uses only earlier comments; the novelty outcome in Table 1 instead uses the completed discussion.

Model development uses the article split and five cross-validation folds. Audience labels resolve relative-vote cutoff ties using the first of ten reproducible draws; editorial labels are observed Editors’ Picks. Each selector has the same number of positive labels within an article.

The conditional-logit model uses article-selector strata and selector-by-feature interactions. XGBoost uses a shared LambdaRank ensemble with a selector indicator. A two-stage hyperparameter search selects models by mean article-level nDCG@ k across selectors and development folds, where k is the number of selected comments in the article. The scalar model has 225 depth-six trees; the text-enabled model has 291 depth-five trees.

The neural models have separate audience and editor prediction heads. Scalar inputs are projected to 128 dimensions; the text-enabled variant additionally projects BGE-M3 embeddings to 256 dimensions. Each head has a 128-unit hidden layer and dropout 0.10. Training uses pairwise logistic loss with four sampled non-selected comments per selected comment and development-fold early stopping. Final fits use the median fold-selected duration: nine epochs for the scalar model and twelve for the text-enabled model. All models are frozen before held-out scoring.

B Detailed Ordering-Policy Effects

Figure 6 shows FORUM scores for each ordering, reply, and pinning combination at both depths. These policy-specific scores complement the average contrasts in the main text, which can conceal differences between configurations.

Comparison with nDCG

Normalised discounted cumulative gain (nDCG) compares discounted feature values with an ideal ordering (Wang et al. 2013). For feature values x_k and policy order π_p , let

$$z_k = \frac{x_k - \min_{\ell} x_{\ell}}{\max_{\ell} x_{\ell} - \min_{\ell} x_{\ell}}, \quad (6)$$

$$\text{nDCG}_d^p = \frac{\sum_{i=1}^d z_{\pi_p(i)} / \log_2(i+1)}{\sum_{i=1}^d z_{\pi_b(i)} / \log_2(i+1)}.$$

where z_k is min–max transformed relevance for comment k , d is the evaluation depth, $\pi_p(i)$ is the comment at rank i under policy p , and π_b orders comments by descending x_k . We use $d = 10$ and $d = N - 1$, matching the FORUM horizons. Discussions with constant feature values are excluded from the corresponding comparison because the relevance transformation is undefined. When a policy hides comments, nDCG preserves the visible prefix and appends omitted comments in ranked-root order, retaining the stored parent-before-child traversal within each root. This differs from FORUM’s linear continuation for hidden material.

Agreement across depths and measures

Across the 84 substantive policy bundles and ten primary outcomes, top-10 and full-discussion FORUM correlate at $r = 0.773$ and Spearman $\rho = 0.811$ (Fig. 7). Feature-specific Spearman correlations are lower for positive sentiment (0.476) and prior reception balance (0.372) than for article similarity (0.894) and lexical diversity (0.877). Early prominence is therefore not interchangeable with full-list presentation—topical similarity and lexical diversity are relatively stable across horizons, whereas positive sentiment and prior reception balance are more horizon-dependent.

Across policy–feature combinations, FORUM and nDCG correlate more strongly at top 10 ($r = 0.901$, $\rho = 0.895$) than at full depth ($r = 0.262$, $\rho = 0.538$; Fig. 8). Full-depth agreement remains feature-dependent: Spearman correlations are 0.644 for positive sentiment, 0.724 for prior reception balance, and 0.975 for article similarity. The measures answer different questions. nDCG discounts lower ranks and measures recovery of ideal feature values; FORUM averages signed, normalised deviations from the expected random trajectory.

C Topic-Model Selection

The topic-model search uses a fixed mixed corpus of 48,887 article passages and 48,887 sampled comments. It evaluates 60 combinations of minimum cluster size, UMAP neighbourhood size, and minimum samples, with three seeds (2025–2027), for 180 fits. Figure 9 reports article-passage and sampled-comment assignment coverage and the median pooled adjusted mutual information (AMI) among documents assigned in both runs of each seed pair. This stability measure is conditional and does not indicate that every document receives a stable topic.

The selected setting (minimum cluster size 15, ten neighbours, minimum samples 5, seed 2026) lies on the coverage–stability frontier: no other setting improves one of the three diagnostics without worsening another. Its median pooled common Adjusted Mutual Information (AMI) is 0.669, with raw fit-assignment coverage of 58.60% for article passages and 33.56% for sampled comments. The qualitative choice also considered top topic coherence and topic count, the number of documents in very small topics, and relative improvement across these diagnostics. The selected setting has a median 719 pre-reduction topics and zero pooled documents in topics of size at most 5 or 10. A neighbouring frontier setting (minimum cluster size 10, 15 neighbours, minimum samples 3) has similar coverage (59.88% and 33.56%) and AMI (0.666), but 1,255 topics and 1,160 pooled documents in topics of size at most 10. The final 439-topic model and its assignment coverage are reported in the “Measuring the presented topic agenda” subsection. Its coverage of 53.59% for article passages and 26.64% for candidate comments comes from applying the final model to the full article and comment corpus.

D Development Vote-Activity Weights

The reconstructed development presentations contain 1,189,267 candidate comments across 2,553 discussions.

Features (41 scalar inputs in total)	Operationalisation
Text and language (7): Comment length; positive and negative sentiment; maximum toxicity; length-adjusted lexical diversity; length-adjusted reading difficulty; URL present	$\log(1 + \text{word count})$; sentiment probabilities from the CardiffNLP XLM-R classifier; maximum chunk toxicity probability from the TextDetox XLM-R classifier; $\text{CTTR} = V/\sqrt{2W}$ and German $\text{SMOG} = \sqrt{30P/S} - 2$ (distinct tokens V , words W , polysyllabic words P , sentences S), each residualised against log length with a six-knot cubic spline and ridge penalty; binary URL indicator.
Discussion context and timing (7): Article similarity; novelty; hours since publication; discussion pace; overnight, weekday shoulder/evening, and weekend daytime/evening	Mean cosine similarity to the three most similar article passages; one minus the maximum cosine similarity to strictly earlier eligible comments (roots for the root scope, all comments for the primary all scope); $\log(1 + \text{hours})$; $\log[(\text{prior comments} + 0.5)/(\text{hours} + 0.5)]$; Vienna-local posting-period indicators: 00:00–05:59 every day; weekdays 06:00–08:59 or 18:00–23:59; weekends 06:00–23:59. The reference is weekdays 09:00–17:59.
Author history (4): Author comments in the preceding 30 days; prior upvote reception; prior downvote reception; author’s earlier comments in the article	$\log(1 + x)$ for prior 30-day and earlier within-article comment counts; $\log[(v + 0.5)/(n + 0.5)]$ for prior upvote/downvote reception, where v is snapshot votes on the author’s n earlier comments in the 30-day window. History is ordered by timestamp.
Reply structure (3): Earlier reply-to-root composition; reply rather than root; centred reply depth	$\log[(\text{prior comments} - \text{prior roots} + 0.5)/(\text{prior roots} + 0.5)]$; binary reply indicator; $\log(1 + \text{depth})$ centred at the development-reply mean and multiplied by the reply indicator so roots have zero centred depth.
AQuA dimensions (20): Relevance; factual claim; opinion; justification; solution proposal; additional knowledge; question; references to users, media, comment content, and personal details; references to format; polite address; respect; screaming; vulgarity; insult; sarcasm; discrimination; storytelling	Twenty expected ordinal scores from the frozen multilingual AQuA adapters, each on its raw 0–3 expected-score scale. The scores are uncalibrated model outputs; hard labels and the weighted composite are descriptive or sensitivity outputs and are excluded from the primary models.
Text representation: Fixed BGE-M3 comment embedding (1,024 dimensions)	L2-normalised comment embedding, precomputed before selection-model training and held fixed. This block is appended only to XGB-T and NN-T; it is not counted among the 41 scalar predictors.

Table 2: Features and their Operationalisation. Continuous predictors are standardised within development fitting data; binary indicators remain 0–1.

Pinning hides 146,950 descendants, leaving 1,042,317 visible comments and 7,634,736 recorded votes. No zero-vote discussion is excluded. Within each discussion, vote fractions sum to one and absent ranks contribute zero. Conditioning the fit on visible length separates rank weighting from the declining number of discussions that reach later ranks.

The log-weight spline described in the “Measuring the presented topic agenda” subsection uses denser knots near the start of the list. Nested development-fold validation selects a second-difference penalty of 10^{-4} . Mean per-story validation cross entropy is 5.411 nats for the selected spline, versus 5.507 for a fitted power law and 5.606 for a fitted exponential. The latter curves are development-fit comparators in Figure 10; presented agenda analyses use the spline only, though these alternatives give qualitatively similar results.

E Asset Licences

The following records identify the upstream licences of the pretrained assets used for feature extraction (checked 16 September 2026).

- **BGE-M3:** BAAI/bge-m3 is released under the MIT licence, as declared in its model card: <https://huggingface.co/BAAI/bge-m3>.
- **Sentiment:** the official XLM-T repository associated with `cardiffnlp/twitter-xlm-roberta-base-sentiment` carries the Apache License 2.0: <https://github.com/cardiffnlp/xlm-t/blob/main/LICENSE>.
- **AQuA:** the upstream repository containing the deliberative-quality code and trained adapters carries

the Apache License 2.0: <https://github.com/mabehrendt/AQuA>.

- **Toxicity:** `textdetox/xlmr-large-toxicity-classifier-v2` declares **OpenRAIL++** (`openrail++`) in its model card: <https://huggingface.co/textdetox/xlmr-large-toxicity-classifier-v2>.

F Compute Resources

The computational experiments ran on a Linux workstation with an AMD Ryzen 9 9950X processor (16 physical cores, 32 hardware threads) and 64 GB installed system memory (approximately 60 GiB visible to the operating system). GPU workloads used one NVIDIA RTX A5000 with 24 GB VRAM.

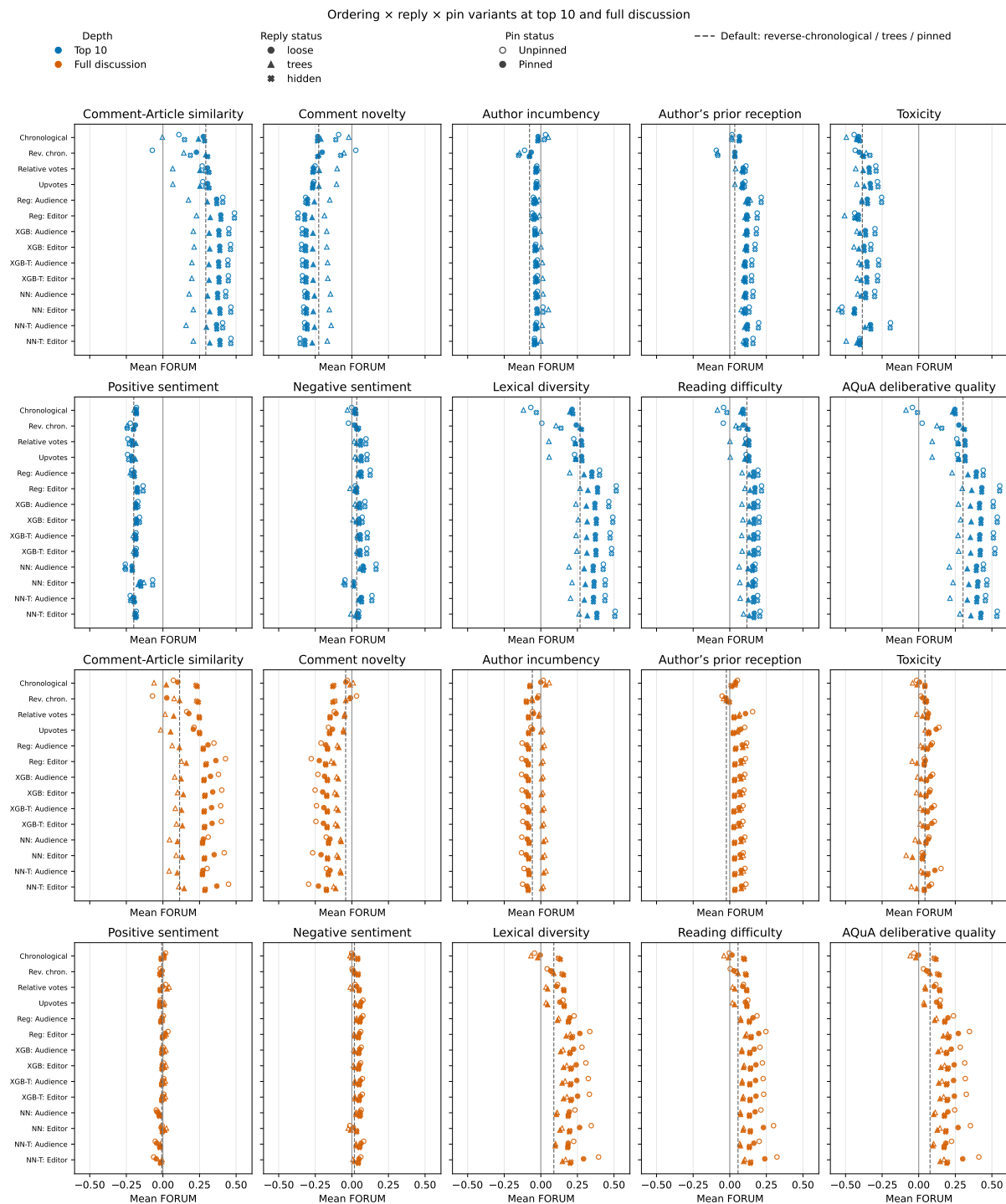


Figure 6: Detailed ordering-policy variants at top-10 and full-discussion depths, showing the six reply/pinning configurations across the ten primary feature outcomes. Positive FORUM values indicate prominence above the expected random trajectory and negative values indicate suppression. Dashed grey lines mark the complete default presentation (reverse-chronological ordering, reply trees, and pinned Editors' Picks). These are feature-specific outcomes, not a single quality ranking.

Top-10 versus full-list FORUM across ranking conditions

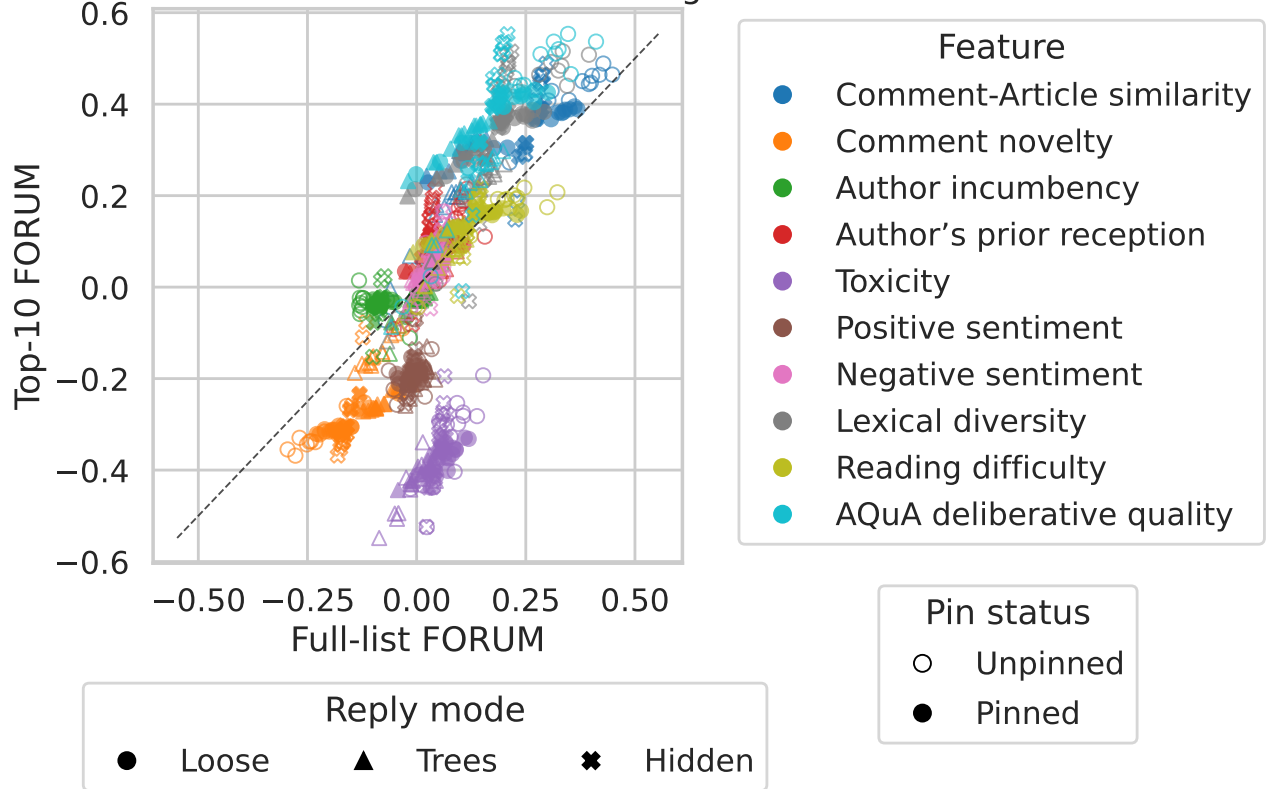


Figure 7: Top-10 versus full-discussion FORUM across ranking conditions. Each point is one of the 84 substantive policy bundles paired with one of the ten primary outcomes; the horizontal axis is full-list FORUM and the vertical axis is top-10 FORUM. Colour identifies the feature, marker shape reply mode, and fill pinning. The dashed line marks equality.

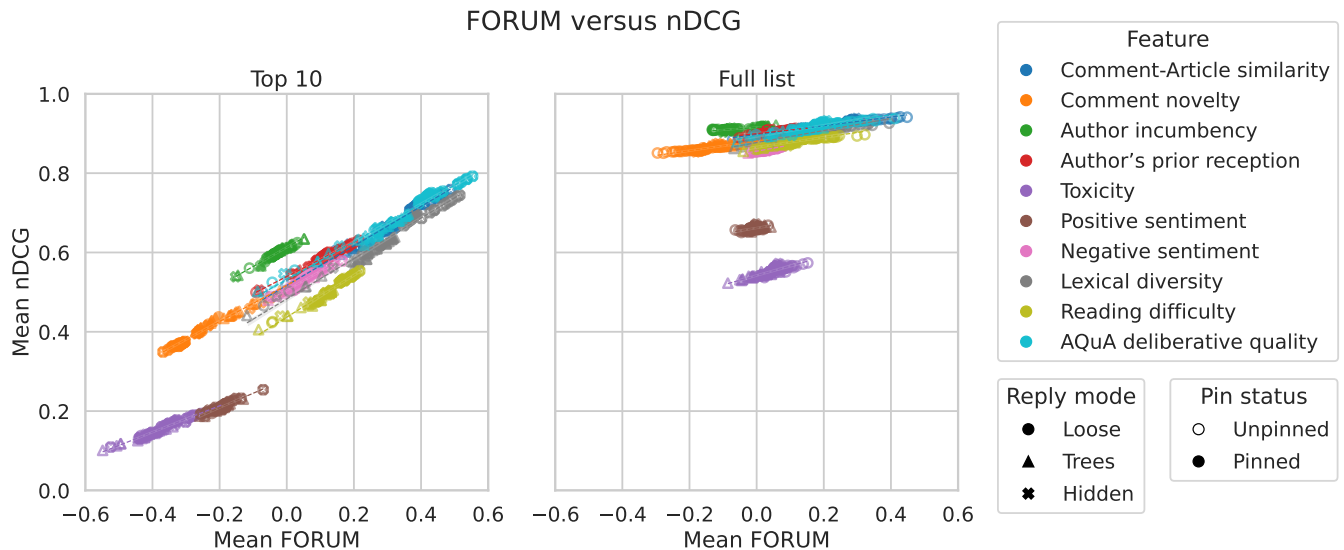


Figure 8: FORUM versus direct nDCG by depth. Each point is one of the 84 substantive policy bundles paired with one of the ten primary outcomes; panels separate top-10 and full-list values. FORUM is on the horizontal axis and nDCG on the vertical axis. Colour identifies the feature, marker shape reply mode, and fill pinning.

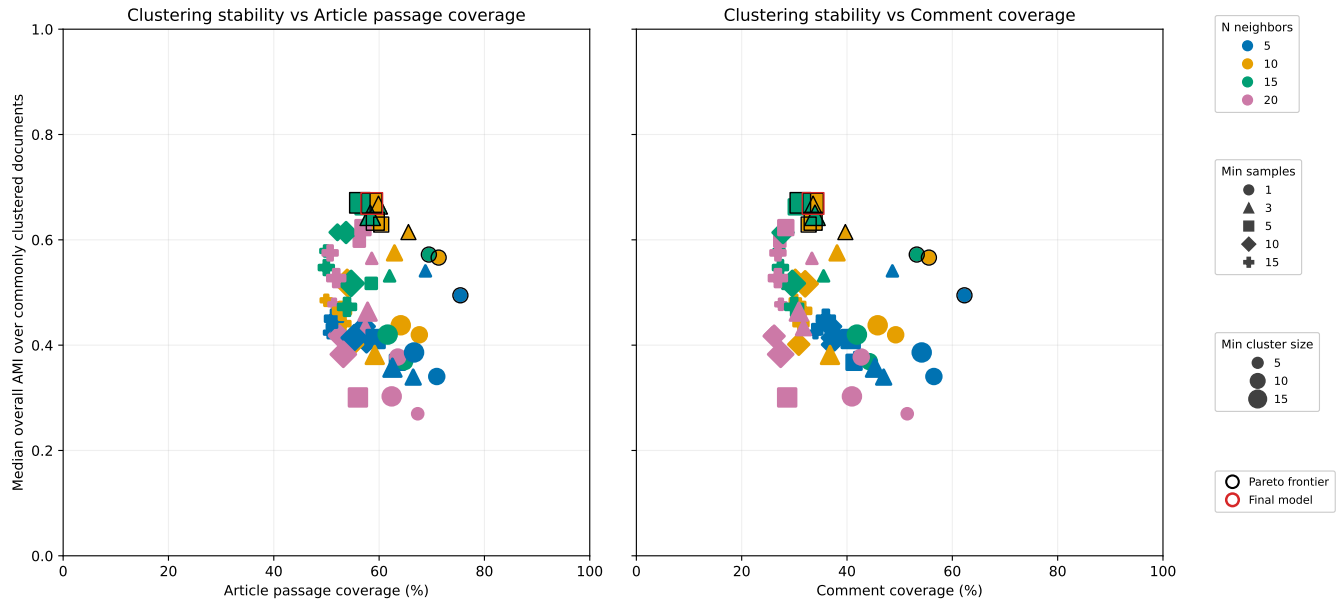


Figure 9: Topic clustering stability versus article and sampled-comment coverage. The two panels share median pooled common-assignment AMI and distinguish the two coverage denominators. Open markers identify the three-objective Pareto frontier; the red outline marks the selected setting.

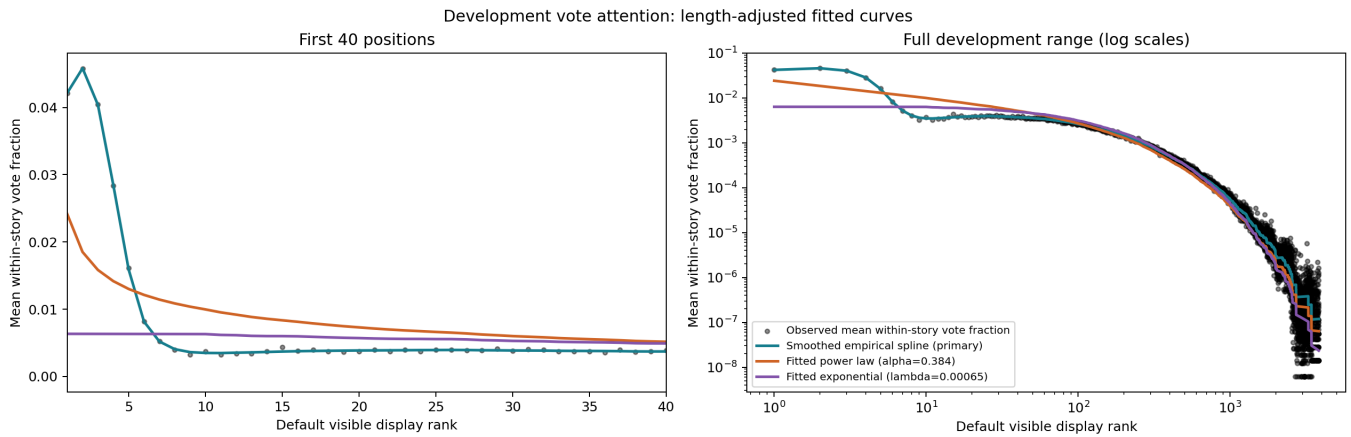


Figure 10: Development vote activity and length-adjusted fitted curves. The first forty positions are shown on linear axes and the full range on log-log axes. Empirical values are equal-discussion mean within-story vote fractions; fitted curves account for visible discussion length. The spline supplies the agenda-analysis weights.